

An Augmented Reality Experience for Climate Justice: Using Spatial Animation to Enhance Perceived Togetherness

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(a)



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Figure 1: An AR Experience for Climate Justice.

Abstract

This project focuses on designing an Augmented Reality (AR) application for climate justice communication. It examines ways to use spatial graphics/animation/sound for multi-user simulations with strategies regarding economic and environmental conflicts.

Keywords

Augmented Reality, Climate Justice, Multi-User Experience, Perceived Togetherness

ACM Reference Format:

Ching-Hua Chuan, Wan-Hsiu Tsai, and Xueer Xia. 2024. An Augmented Reality Experience for Climate Justice: Using Spatial Animation to Enhance Perceived Togetherness. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697964>

1 Introduction

Understanding climate injustice—the unequal distribution of adverse climate impacts that disproportionately affects the more vulnerable—is crucial for facilitating climate agreements among different sectors [Roosvall and Tegelberg 2020]. However, due to the vastness and complexity of historical and territorial inequalities in climate impacts, facilitating understanding of climate justice

among the public remains a key challenge. Research suggests that incomprehension of environmental connectivity can make people perceive climate impacts on other regions as distant and irrelevant, diminishing the sense of shared responsibility [Smetanová et al. 2018].

This research focuses on designing, implementing, and evaluating an AR application for climate justice communication. Research has highlighted the effectiveness of immersive technologies including AR and virtual reality (VR) for enhancing engagement with social issues. For example, Schutte and Stilinović [Schutte and Stilinović 2017] showed that experiencing a documentary about a refugee camp in a VR environment elicits empathy. Wang et al.'s [Wang et al. 2021] AR mobile app was effective for promoting pro-environmental behavioral changes. This study thus focuses on AR that enhances individual users' own physical space with virtual content, which can be particularly effective for communicating concepts that have strong physical references [Butts and Jones 2021].

2 Designing Climate Justice AR Experience

2.1 Spatial Settings

In this AR experience, three participants will interact with a shared virtual environment while physically being in the same space. The virtual environment consists of three adjacent territories representing areas with different economic resources and climate vulnerabilities. Participants will be assigned to one of the territories based on the outcome of a spinning wheel and given several options to develop their own regions. Some options are more financially costly but more ecologically sustainable while others can quickly boost the economy but with greater negative environmental impacts. The

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SA Posters '24, December 03–06, 2024, Tokyo, Japan
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ACM ISBN 979-8-4007-1138-1/24/12
<https://doi.org/10.1145/3681756.3697964>

economic index and environmental index of each territory will be displayed and dynamically updated during the entire experience. More development options become available as the economic index increases, but an area-specific natural disaster (e.g., flood at the coastal city) occurs when the environmental index is too low. Once a natural disaster strikes, causing damage in all three areas, the economic index of each area will be dramatically reduced. Participants will witness the consequences of their actions on each other (e.g., deforestation of rainforest leads to mudslides and wildfires in other areas). This shared AR experience provides numerous animations and visualization that demonstrate the conflicts between economic development and environment conservation, and more importantly, the environmental connections among all territories.

2.2 Challenges

There exist many challenges in designing this multi-person AR experience regarding animation space and timing. A country may experience more environmental damages due to its natural condition, such as geographical location, although it produces significantly less pollution compared to other nations. To demonstrate such environmental injustice, it is crucial to provide realistic settings to illustrate different environmental conditions. When users are building their own areas, the animations illustrating different development options must be clear to each user without distracting or interrupting others. Additionally, each user's actions must be obvious and easily observable for other users to notice what this particular user has done. The animations of natural disasters must be dynamically displayed with sound effects, starting in one region and spreading to other areas to create damage. Multiple natural disasters can also occur simultaneously. This project thus focuses on controlling animation timing and using spatial sound to manage multiple moving animations while ensuring the AR experience remains easy for users to follow.

2.3 Simulation for User Experience

To determine the parameters in the experience, such as cost of each development option and the starting environmental index for each area, simulations were conducted with random plays for 2000 times to ensure that (1) climate injustice is demonstrated (i.e., poor areas suffer more natural disasters than rich areas), (2) most devastating consequences of natural disasters (i.e., natural disasters significantly reduce a participant's economic index, leaving the participant unable to afford any options) can occur but not too frequent, and (3) the natural disaster occurs in the early stage of the experience to ensure demonstration of economic and environmental conflicts. The simulation results are shown in Figure 2.

3 Pilot Testing

To evaluate the effectiveness of the AR experience, triad interviews were conducted to assess participants' feedback on the design, user experience, and perceived togetherness or "linked fate" [Bejarano et al. 2021] in relation to environmental connectivity. Specifically, the interviews seek to answer the following research questions:

- *RQ1*. How does the 3D graphics impact participants' sense of control and their perceived togetherness?

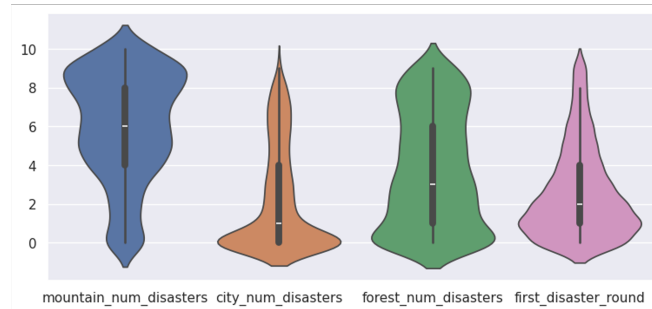


Figure 2: The simulation results for the experience.

- *RQ2*. How does the flow of the animations influence participants' experience and understanding of the design?
- *RQ3*. How would the participants respond to the interconnectedness of their decisions in the virtual territories?

A pilot test with three participants was conducted. The experience was repeated three times in the test, and each time participants were randomly assigned to a territory. In the first round, participants were mostly trying to figure out how the experience works. Once they understood the design, they attempted to gain the most economic credits in the second round, which quickly resulted in the "unjust" condition. Surprisingly, participants collaboratively figured out a strategy together in the third round, taking turns to let others increase their economic index while keeping the environmental index above the threshold.

In terms of the design, participants felt that the 3D graphics and animations were clear and attractive. While they were operating their own devices in the experience, they naturally talked to each other and discussed their actions and consequences. Although that the natural disasters successfully grabbed participants' attention, the sound effect was not well perceived because the application was running on a head-held device. In the future, we plan to conduct more tests and expand this experience for mixed reality headsets to provide an even more immersive engagement.

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